

Reward Distribution Estimation

Point Estimation

$$\hat{p}_\tau^{\text{pt}} = \frac{1}{N} \sum_{i=1}^N X_{\tau,i}$$

$$\text{Bias}(\hat{p}_\tau^{\text{pt}}) = 0 \quad \text{Var}(\hat{p}_\tau^{\text{pt}}) = \frac{p_\tau(1-p_\tau)}{N}$$

Discounted Beta-Bernoulli Estimation

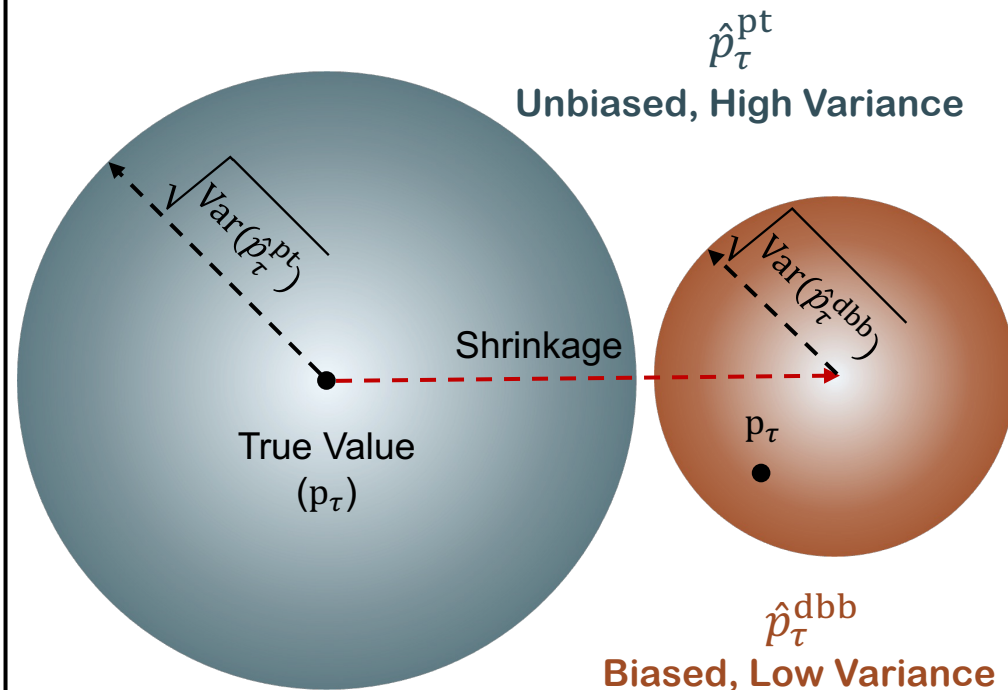
$$\alpha_\tau = \lambda \alpha_{\tau-1} + S_\tau \quad \beta_\tau = \lambda \beta_{\tau-1} + N - S_\tau$$

$$\hat{p}_\tau^{\text{dbb}} = \frac{\alpha_\tau}{\alpha_\tau + \beta_\tau}$$

$$\text{Bias}(\hat{p}_\tau^{\text{dbb}}) \neq 0 \quad \text{Var}(\hat{p}_\tau^{\text{dbb}}) \ll \text{Var}(\hat{p}_\tau^{\text{pt}})$$

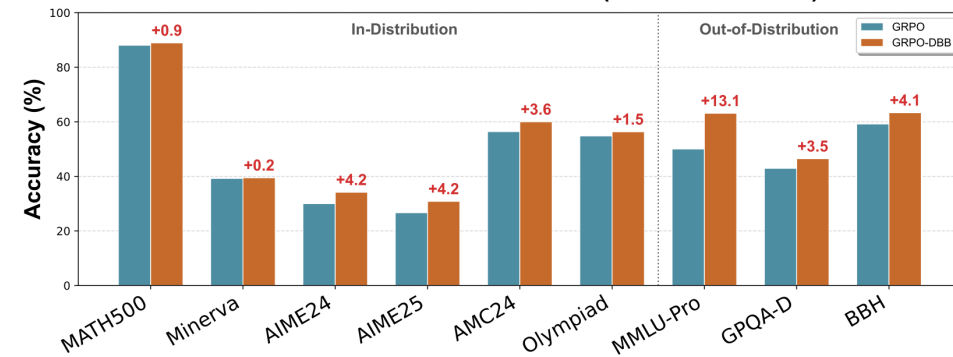
Bias-Variance Trade-off & Shrinkage

$$\text{MSE}(\hat{p}_\tau^{\text{dbb}}) < \text{MSE}(\hat{p}_\tau^{\text{pt}})$$



Performance Comparison

Qwen3-8B-Base Performance (DAPO-Math-17k)



Qwen3-1.7B-Base Performance (DAPO-Math-17k)

